MILITARY SPECIFICATION

KIT, FILLING FOR SILVER-ZINC BATTERY 16XPM-05

1. SCOPE

1.1 This specification covers special requirements, quality assurance provisions, and the preparation for delivery criteria for one type of filling kit for silver-zinc battery 16 XPM-05 used on the Atomic Demolition Charge.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of

DRAWINGS

ORDNANCE CORPS

8808362	-	Packing and Marking for Storage and Shipments.
8861697	-	Bag, Packing Ammunition, for Battery Filler Kit.
9207938	-	Filling Kit and Instructions for Silver-Zinc Battery 16 XPM-05.
EL-9207938	-	Equipment List.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

FSC 1195

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3.2 Assembly. The filler kit shall comply with all requirements specified on Drawing (dwg) 9207938, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 Low temperature. Each filler kit shall withstand -65 ± 5 degrees Fahrenheit (°F) without evidence of damage, cracking, or leaking and each electrolyte bottle shall comply with 4.3.3.1 after the filling kit has returned to ambient temperature. Testing shall be as specified in 4.4.2.

3.4 Workmanship.

3.4.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.4.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.4.3 Burrs. Parts shall be free from burrs.

3.4.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.4.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. Not applicable.

4.3 Product acceptance inspection.

4.3.1 Lot formation. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

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4.3.1.1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies, or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable classification. Any characteristic in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table I
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table II
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics. Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Filling kit prior to sealing (see dwg 9207938).

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major:		
	101. No bottle missing or improperly sealed	Visual
	102. Correct amount of electrolyte fluid in the bottle	Visual
	103. No component missing or damaged	Visual
Minor:		
	201. Workmanship (see 3.4)	Visual/Manual

4.3.2.1.2 Filling kit (see dwg 9207938).

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major:		
	101. Marking	Visual
	102. No box damage	Visual
Minor:		
	201. Workmanship (see 3.4)	Visual/Manual

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4.3.2.1.3 Bag, packing, ammunition, for battery filler kit (see dwg 8861697 and section 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major:		
101.	Boxes secured with tape	Visual
102.	Air exhausted from bag	Visual
103.	Proper heat seal	Visual/Manual
104.	Objective evidence (see 6.2) for heat seal test in accordance with MIL-P-116	Visual
105.	Marking	Visual
106.	Packaging materials (see 3.1)	Visual
Minor:	None defined	

4.3.2.1.4 Packing and marking for storage and shipments (see dwg 8808362 and section 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major:		
101.	Gross load weight, max	Measure*
102.	Contents tight in package	Manual
103.	Correctly packed	Visual/Manual
104.	Package material (see 3.1)	Visual
Minor:		
201.	Marking	Visual
202.	Workmanship (see 3.4)	Visual/Manual

*Determine weight with a suitable scale.

4.3.3 Testing. All tests shall be performed. Tests need not be performed in the order specified.

Test	Test Classification
a. Leakage	Major
b. Low temperature	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Leakage. Each electrolyte bottle shall be tested in accordance with 4.4.1. If any item fails to comply with drawing requirements, the item shall be rejected and removed from the lot.

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4.3.3.2 Low temperature (see 3.2). Each filler kit shall be tested in accordance with 4.4.2. If any item fails to comply with 3.2, the item shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (See EL-9207938). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment. The contractor shall obtain approval of such designs, operating instructions, and special calibration instructions as specified in 6.3 prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired.
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL.

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accurancy tolerances of the test equipment used.

4.4.1 Leakage (see 4.3.3.1). Each bottle shall be filled and sealed. The bottles shall be tested in accordance with dwg 9207938. Any indication of leakage shall reject the bottle.

4.4.2 Low temperature (see 3.2). Each filler kit shall be placed in a chamber at -65°+5°F for a minimum of 8 hours. After returning to ambient temperature without the use of forced heating the bottles shall comply with 4.3.3.1.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The filler kit shall be packaged in accordance with dwgs 8861697 and 8808362 using the specified moisture barrier.

5.1.2 Level C. The filler kit shall be packaged in accordance with dwg 8808362 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The filler kit shall be packed in accordance with dwgs 8861697 and 8808362.

5.3 Marking. Marking shall be as specified in accordance with dwgs 8861697 and 8808362.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment. The contractor shall obtain approval of equipment designs, operating instructions, and calibration instructions from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-NR4.

Custodian:
Army - MU

Preparing activity:
Army - MU

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